

ES6 imports and 'is not a constructor' in Jest.mock

Asked 1 year, 2 months ago · Active 1 year, 2 months ago · Viewed 873 times



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Similar to [Jest TypeError: is not a constructor in Jest.mock](#), except I am using ES6 imports - and the answer given to that question is not working on my situation.

Following [the Jest .mock\(\) documentation](#) I am attempting to mock the constructor `Client` from the `pg` module.

I have a constructor, `Client`, imported from an ES6 module called `pg`. Instances of `Client` should have a `query` method.

```
import { Client } from "pg";
new Client({ connectionString: 'postgresql://postgres:postgres@localhost:5432/database'
});

export async function doThing(client): Promise<string[]> {
  var first = await client.query('wooo')
  var second = await client.query('wooo')
  return [first, second]
}
```

Here's my `__tests__/test.ts`

```
const log = console.log.bind(console)

jest.mock("pg", () => {
  return {
    query: jest
      .fn()
      .mockReturnValueOnce('one')
      .mockReturnValueOnce('two'),
  };
});

import { Client } from "pg";
import { doThing } from "../index";

it("works", async () => {
  let client = new Client({});
  var result = await doThing(client);
  expect(result).toBe(['one', 'two'])
});
```

This is similar to the answer given in [Jest TypeError: is not a constructor in Jest.mock](#), but it's failing here.

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```
const mockDbClient = new Client({ connectionString: env.DATABASE_URL });
```

fails with:

```
TypeError: pg_1.Client is not a constructor
```

I note [the docs mention](#) `esModule: true` [is required when using default exports](#), but `Client` is not a default export from `pg` (I've checked).

How can I make the constructor work properly?

Some additional notes after getting an answer

Here's a slightly longer-form version of the answer, with comments about what's happening on each line - I hope people reading this find it useful!

```
jest.mock("pg", () => {  
  // Return the fake constructor function we are importing  
  return {  
    Client: jest.fn().mockImplementation(() => {  
      // The constructor function returns various fake methods  
      return {  
        query: jest.fn()  
          .mockReturnValueOnce(firstResponse)  
          .mockReturnValueOnce(secondResponse),  
        connect: jest.fn()  
      }  
    })  
  }  
})
```

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76 ● 324 ● 402

1 Answer

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When you mock the module, it needs to have the same *shape* as the actual module. Change:

3

```
jest.mock("pg", () => {  
  return {  
    query: jest  
      .fn()
```

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...to:

```
jest.mock("pg", () => ({
  Client: jest.fn().mockImplementation(() => ({
    query: jest.fn()
      .mockReturnValueOnce('one')
      .mockReturnValueOnce('two')
  }))
}));
```

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edited May 15 '20 at 17:11

answered May 15 '20 at 16:52



Jacob

73.4k ● 23 ● 139 ● 216

- 1 Thanks Jacob! What I learnt was that 1. The mock implementation must be an object with what we're importing as the keys. 2. the Constructor, fairly obviously in retrospect, should be a function. –

[mikemaccana](#) May 15 '20 at 17:17